DESK LIGHTHOUSE · STEP 2: THE SCIENCE AND ENGINEERING

Why the lamp stands still and the brass turns

The working behind every number on the Rev D sheet (Rev C approved 2026-11-09; Rev D, the open lantern without the house, drawn that evening for approval): a fixed lamp and a turning window, how much light leaves it, a filament behind a resistor and why the rail is 5 V, a gearmotor at a third of its voltage, the boost and the budget from the cell, one diode and the rail as the cell empties, charging through the coaster, why the rods lean, the coaster as the fuse, a magnet and a reed, the spine and its wheel.
Read after Desk Lighthouse Build. Nothing measured yet; every figure is derived from datasheet and listing numbers and the sheet's dimensions.

The whole thing in one sentence

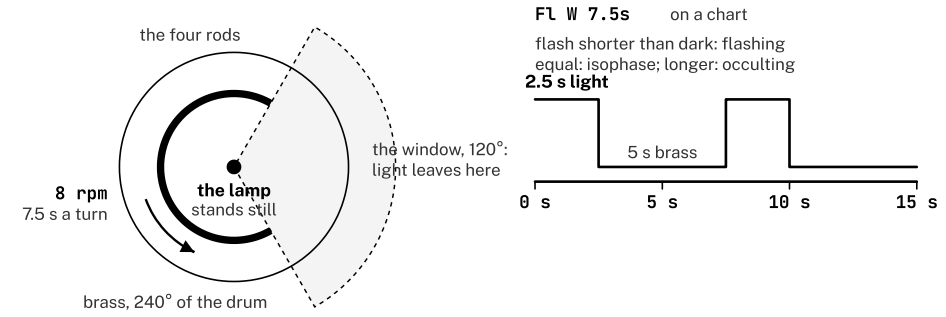
A 26 mm LED filament stands still in the open on four brass rods; a brass shutter with a 120° window turns round it once every 7.5 s from a gearmotor at the foot; a 16340 cell beside the motor feeds a 5 V boost that runs both; and a charger fills the cell through a notch in the coaster.

Everything on the sheet follows from that: why the lamp is fixed and the brass moves, why the rail is 5 V and not the cell, why the motor runs at a third of its voltage, why the rods lean, why a reed and not a sensor, why the coaster is the fuse. This page takes each in the order the sheet made them and shows the working.

A fixed lamp and a turning window: the light's character

A lighthouse does not spin its lamp. The lamp burns still at the focus and the lens turns round it, so the beam sweeps the horizon and nobody has to feed a moving flame. The sheet does the same with a brass drum: the filament is wired to nothing that moves, and the window carries the light round the room.

The pattern a light makes is its character, and sailors tell lights apart by it. Here the window is 120° of the drum, so the room sees light for 2.5 s and brass for 5 s, every 7.5 s. In a chart's notation that is Fl W 7.5s, a flashing light: a flash shorter than the dark between. A 180° window would make it isophase, light and dark equal; wider still, occulting.



A lighthouse burns a fixed lamp inside a turning lens; here the lens is a brass drum with a window. 48° a second: on a wall 2 m away the edge of the light walks at 1.7 m/s.

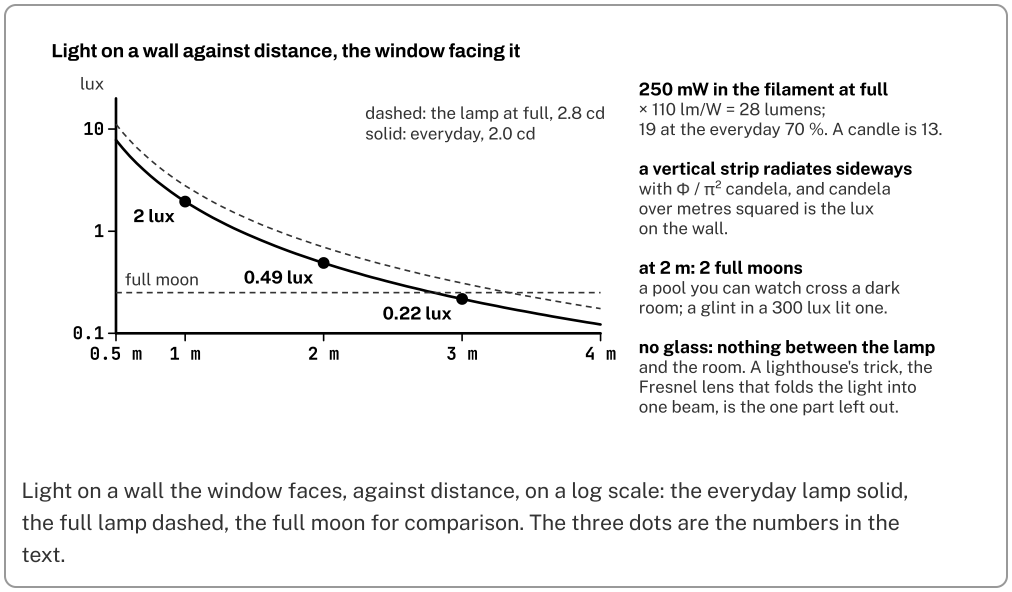
Left: the lamp on the axis, the brass drum round it, the window and the fan of light that leaves it. Right: what the room sees, 2.5 s of light and 5 s of brass, twice.

The speed is 8 rpm, 48° a second. On a wall 2 m away the edge of the light crosses at 1.7 m/s, a slow walking pace: slow enough to watch, quick enough that the eye reads a sweep and not a lamp going on and off. Faster looks like a warning light; slower, like a fault.

How much light leaves the window

The filament takes 250 mW at full, and a filament LED gives about 110 lumens a watt: about 28 lumens, 19 at the everyday 70 %. A candle gives 13. A vertical strip radiates sideways with an intensity of its flux over π^2, so 2.0 candela everyday and 2.8 at full.

Intensity over distance squared is the light on a wall: 2.0 lux at 1 m, 0.49 at 2, 0.22 at 3. A full moon gives 0.25 lux and a lit room 300. In a lit room the sweep is a glint moving along the wall; with the lights down it is 2 moons at 2 m, a pool you can watch cross the room.



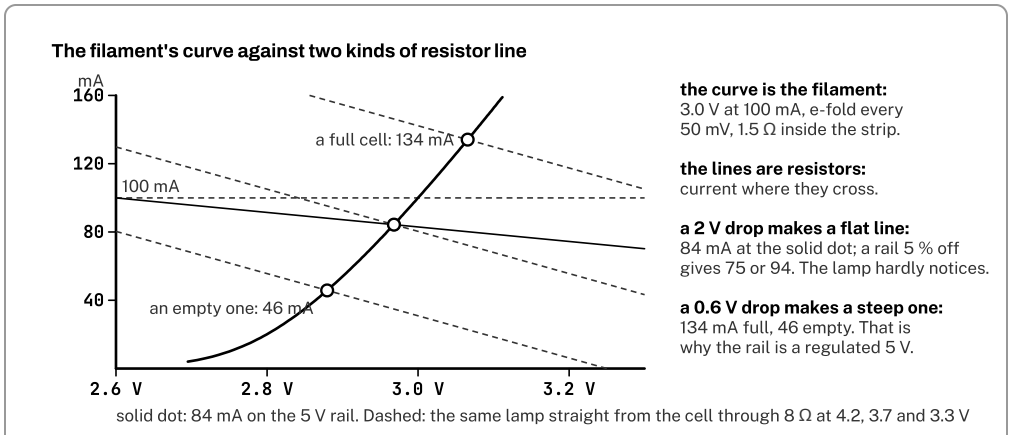
Light on a wall the window faces, against distance, on a log scale: the everyday lamp solid, the full lamp dashed, the full moon for comparison. The three dots are the numbers in the text.

There is no glass. Rev C had an acrylic tube round the lantern and Rev D took it out at the builder's word, and nothing optical went with it: a plain tube is a window, not a lens, since light from the axis meets both its surfaces square on and goes straight through. A lighthouse's trick, the Fresnel lens that folds a lamp's whole output into one beam, is the one part of a lighthouse the sheet leaves out either way. The window lets out a third of the azimuth at a time; the rest hits the brass inside the drum, which reflects about 60 % of warm light, so some of it follows the window out. Heat is not a worry: 163 mW of heat from a strip in free air, inside a brass drum of 21 cm^2 open at both ends, warms the drum at most 8 °C at full, 5 everyday; the air through it takes most of that away.

The filament behind a resistor, and why the rail is 5 V

An LED filament is a row of blue dies side by side on a ceramic strip under a yellow phosphor, which is why it wants a diode's 3 V and not a bulb's 120. A diode's current climbs steeply with the voltage across it, e-fold every 50 mV or so here, so no supply may be connected to it directly: a resistor in series sets the current, and the current is where the filament's curve meets the resistor's line.

From the 5 V rail through 23.5 Ω the two cross at 2.97 V and 84 mA, under the strip's 100 mA rating (the sheet's 83 mA takes the voltage as a flat 3.0). The resistor has 2 V across it, and that is the point: a 5 % wobble in the rail moves the current only to 75 or 94 mA.



The filament's curve against the resistor lines. Solid: from the 5 V rail through 23.5 Ω, a flat line. Dashed: the resistor that would give the same current straight from a 3.7 V cell, at a full cell, a half-full one and an empty one.

Run the same lamp straight from the cell and the resistor shrinks to 8 Ω, because only 0.68 V is left to drop at 3.7 V. Then a full cell pushes 134 mA through it, over the rating, and an empty one 46: bright for the first hour, dim for the rest. The boost's price is 12 % and the resistor's 2 V another 40, so the lamp gets 52 % of what the cell pays.

A constant-current driver fed from the cell would do better: about 37 mA less from the cell with the beacon on, 9.3 hours a charge instead of 6.6. The sheet spends it for one board fewer and a rail the motor wants anyway; it is the upgrade if the hours ever feel short.

Warm white at 2700 K, and the heat

Colour temperature is the colour of a glowing body: a candle at 1 850 K, a tungsten bulb at 2 700, halogen at 3 000, the midday sun at 5 500. The lamp in a lighthouse you remember from a photograph is an incandescent bulb behind a lens, 2 700 to 3 000 K; the oil flames before it were a candle's colour; the LED beacons that replaced them are a colder white. The sheet picks the bulb.

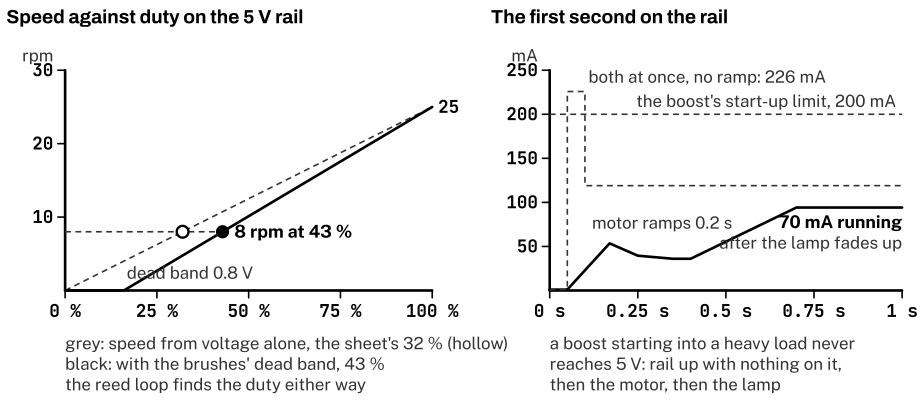
The family's spare filaments are 2 200 K, a candle, and they are right for a pool of amber on brass, the hologram's and the satellite's job. A lighthouse's lamp is the light itself, and the house rule says white or warm white. At 2 700 K the filament reads as a lamp and the brass round it reads as brass, not as more amber. The phosphor is yellow when it is off, the one thing about an LED filament that looks wrong in a lantern; the shutter closes over it.

PWM dims it at 5 kHz, 104 pulses for every degree the shutter turns, so neither the eye nor a phone camera sees anything but the sweep. Dimming by duty keeps the colour: a phosphor LED run at lower current drifts a shade bluer, a pulsed one does not. The filament runs at a quarter of the power each filament carries in a 4 W bulb, in air a couple of degrees over the room; the two resistors put 167 mW into the coaster's hollow and warm the brass 1.6 °C.

A gearmotor at a third of its voltage

A brushed DC motor's speed follows its voltage and its torque follows its current. The N20 is a 30 000 rpm motor behind a 1000:1 gearbox: 30 rpm at 6 V, 25 on the 5 V rail. The sheet wants 8, so the driver chops the rail at 20 kHz and the motor sees the average, 32 % duty on paper.

On paper, because a small motor does not start until about 0.8 V has overcome its brushes and its gears; above that the speed climbs straight. With the dead band the 8 rpm point sits nearer 43 % duty. That is why the reed is there: the firmware measures each revolution and trims the duty until the period is 7.5 s, whatever this motor's batch and bearings do. The regulated rail serves the same loop: a motor fed from the cell would slow as the cell emptied, and the loop would chase it all day.



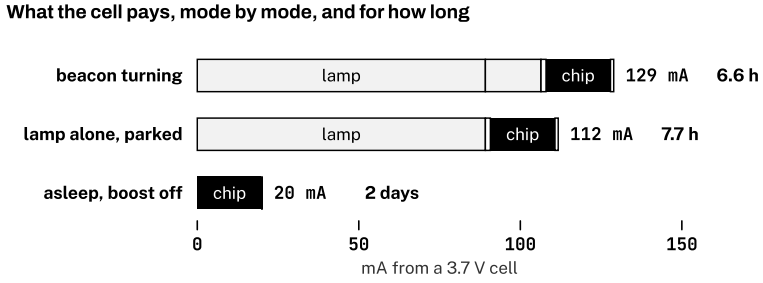
Left: output speed against duty, with and without the brushes' dead band; the sheet's 32 % and the likelier 43 %. Right: the current on the 5 V rail in the first second, ramped as the firmware does it, and the step both loads would make switched on at once as the boost starts.

The gearbox turns speed into torque, 1000 times over; the shutter, a 5 g drum on a spine in a tube-ring hub, needs a thousandth of what the output can give. The danger runs the other way: if the shutter ever jams the loop winds the duty up to 100 %, the stalled motor takes 333 mA from the rail, and plastic gears do not like a stall. The reed's silence catches it in two periods. 20 kHz is above hearing, and the winding's time constant of 67 µs against a 50 µs period smooths the chopped rail into a near-steady current; the gearbox still whispers at 133 Hz, and the hot-glue bed takes most of it.

The ramp is for the boost. While a boost converter starts it can feed only about 200 mA; more at that moment holds the rail down for good. So the rail comes up with both outputs at zero, the motor's duty ramps over 0.2 s and the lamp fades up after it: nothing ever lands on the rail as a step. Without the ramp a stalled rotor at 143 mA and a cold lamp at 83 would be 226 mA in one step. The brake is a short: both inputs high, the motor's own back-EMF stops the rotor in about 30 ms, 1.4° of the shutter.

The boost, and the budget from the cell

A boost converter stores energy in an inductor a million times a second and lets it out at a higher voltage; this one makes 5 V from the cell's 3.3 to 4.2 at about 88 %. Power in is power out over the efficiency, so every milliamp on the 5 V rail costs 1.54 from a 3.7 V cell. The sheet's rail of 70 mA is 108 from the cell; the controller takes its 20 straight from the cell through the diode, and the boost keeps about 1 for itself: 129 mA with the beacon turning.



The cell's current in the three modes, split by what spends it, with the hours that follow from 90 % of 950 mAh.

The lamp is 69 % of it. The motor, at a third of its voltage and a third of the time, is almost nothing. The controller awake is the second biggest line, and when the boost is off it is 100 % of the 20 mA a sleeping lighthouse spends. Light sleep on the

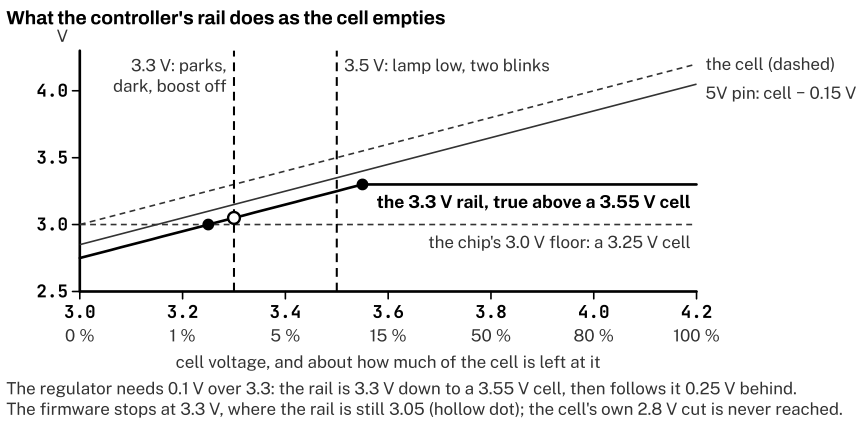
controller, the satellite's trick, would take that figure to under 1 mA and the 2 days to months; it is firmware, not wiring, and the sheet leaves it for the bench.

GPIO7 is the other half of the budget. The boost's enable pin shuts it down with a true disconnect, so a dark, parked lighthouse has no 5 V rail at all and the driver draws nothing. The pin is pulled up to the cell on the board, so a controller that has crashed would leave the rail on; the firmware drives it low first thing at boot and never leaves it floating.

The cell, one diode, and the rail as the cell empties

The cell is the family's Nitecore 16340: 16.6 × 34.1 mm, 950 mAh, 3.6 V nominal, 4.2 full, with its protection under the wrap and a button top. A lithium cell's voltage falls slowly from 4.2 to about 3.5 over most of its charge, then quickly; the firmware's 3.3 V cut-off leaves a few per cent in it, which is the 90 % the hours count on. The protection's own cut at 2.8 V is a last line it never reaches.

The SuperMini has no battery input. Its 5V pin is the USB socket's VBUS with nothing in between, so a flashing cable would put 5 V on the cell; one Schottky between the charger's BAT pad and the pin, band toward the controller, drops 0.15 V and lets nothing back. Behind the pin a small regulator makes 3.3 V and needs about 0.1 V of headroom, so the rail is a true 3.3 V down to a 3.55 V cell and follows the cell down below that: 3.05 V at the cut-off, still over the chip's 3.0 V floor, which a 3.25 V cell would reach. The cut-off sits where the controller still works properly, not where the cell dies.



The cell, the 5V pin and the rail against the cell's voltage, with about how much of the cell is left at each. The rail is a true 3.3 V down to 3.55 V, which is most of the cell; the firmware's two lines are at 3.5 and 3.3 V.

Two 100 kΩ halve the cell for GPIO4: 2.10 V at full, inside the ADC's range at its 11 dB setting, 21 µA for ever, 5 years of the cell. The C3's ADC is noisy and its scale is per chip, so the firmware averages a burst of readings once a minute and the first bench run writes one calibration point into flash; a 100 nF across the lower resistor would quiet it further if the readings jitter. Below 3.5 V the lamp dims and blinks twice; below 3.3 the shutter parks, the lamp goes out, the boost goes off and the controller waits for the button or a cable.

Charging through the coaster

The charger is a linear chip, a controlled resistor between the cable's 5 V and the cell: 500 mA with its jumper cut, 0.53 C for this cell, what a 16340 charger feeds it. Constant current puts about 80 % in over 1.5 h; then it holds 4.2 V and the current tapers for 0.8 h more, 2.3 h in all. Its safety timer stops at 6 h, 2.6 times that, so a healthy charge never meets it.

The drop across a linear charger is heat: 1.3 V × 500 mA = 650 mW in a chip glued up to the brass ceiling of the hollow. The coaster is its heatsink, 107 cm^2 of brass, and it warms about 6 °C while the cell fills: a warm coaster is how you know it is charging, besides the orange LED in the notch. The chip throttles itself at a 125 °C junction and never gets near it.

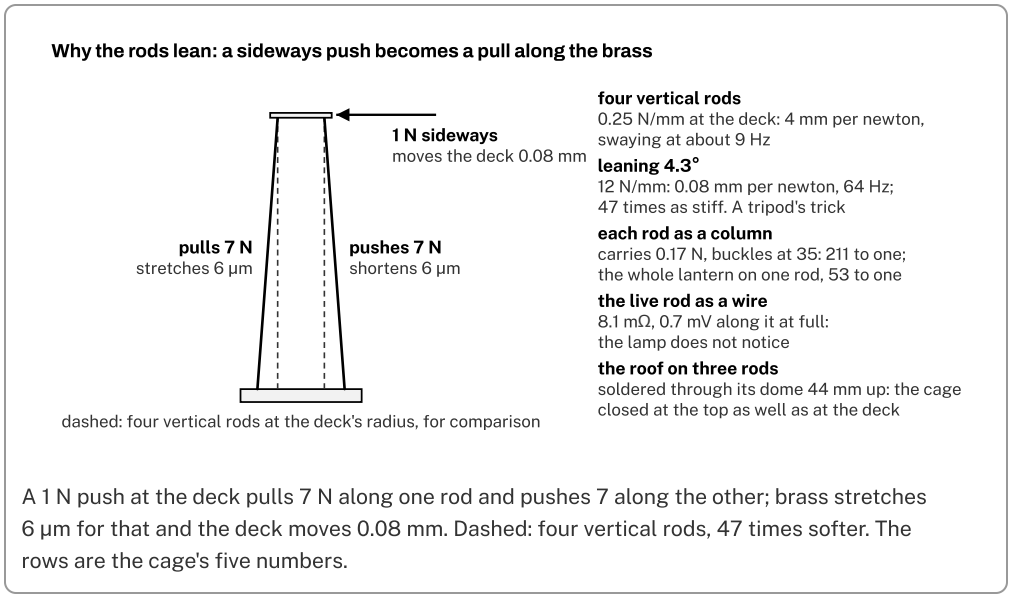
Power path means the cable feeds the loads first and the cell gets the rest, so the lighthouse runs while it charges and the cell is never run down by its own lamp with a cable in. Charging and running together ask the adapter for about 0.6 A; any 1

A phone adapter does. The cell's own cost is calendar life: a 16340 kept at 4.2 V ages faster than one that is cycled, so a lighthouse that lives on its cable should be charged and unplugged rather than left charging; the chip's timer stops it after six hours either way.

Why the rods lean: four wires as a tower

Each rod is a 1.5 mm brass column 168 mm long between the coaster and the deck. Fixed at both ends it buckles at 35 N and carries 0.17, a quarter of the lantern's 67 g: 211 to one. The whole lantern on one rod would still leave 53. Buckling is not the question; sway is.

Four vertical rods would sway like four springs. A rod's bending stiffness goes as the fourth power of its diameter over the cube of its length, and four of these give 0.25 N/mm at the deck: 4 mm for every newton, swinging at about 9 Hz. Lean them 4.3° and a sideways push has to stretch two rods and shorten two, which brass resists along its length and not across it: 12 N/mm more, 47 times as stiff, 0.08 mm per newton, 64 Hz. A camera tripod stands on the same idea. The lean is why the cage looks like a lighthouse, and why it feels like one piece.



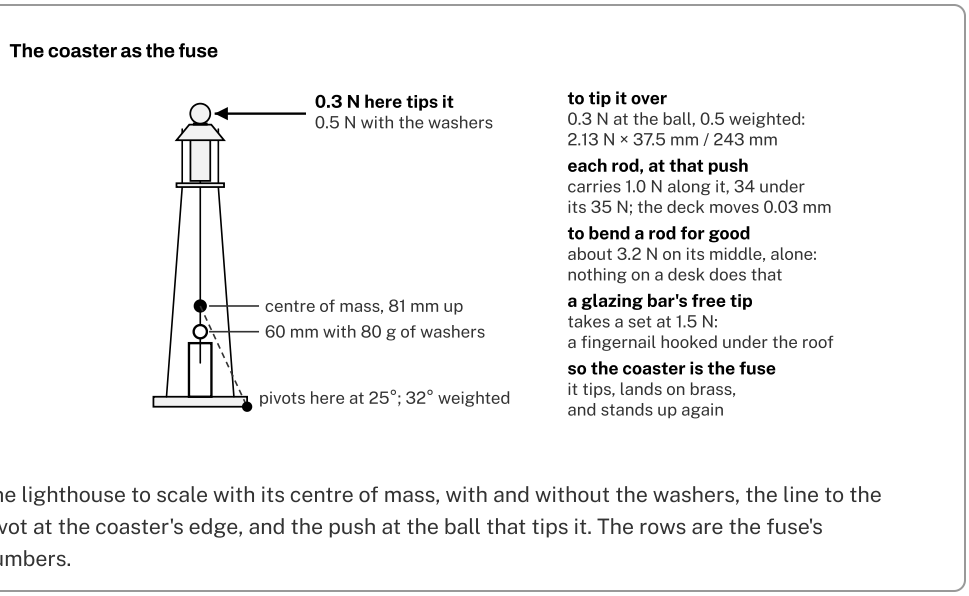
A 1 N push at the deck pulls 7 N along one rod and pushes 7 along the other; brass stretches 6 µm for that and the deck moves 0.08 mm. Dashed: four vertical rods, 47 times softer. The rows are the cage's five numbers.

The deck is what makes it a frame: a solid brass disc threaded on all four rods and glued, so no rod bends alone. Above it the rods carry nothing but their own 44 mm as glazing bars. Two of them are wires as well, 8.1 mΩ each, 0.7 mV along the live one at full, which the lamp does not notice. Above the deck three of the rods pass through the roof's dome and are soldered flush from outside, so the cage is closed at the top as well as at the deck and the roof cannot rock; the live rod stops 42 mm up, booted, and never touches it.

The coaster as the fuse

217 g stand 243 mm tall with their centre of mass 81 mm up: 80 g of it in the coaster and 26 in the cell and its tray low at the foot. The centre of mass over the coaster's 37.5 mm radius gives the tipping angle, 25°, and a push of 0.3 N at the ball tips it. 80 g of steel washers in the hollow bring the centre down to 60 mm, 32° and 0.5 N: a brush of a sleeve rocks it, a hand knocks it over.

Knocking it over is what it does instead of bending. Under the tipping push each rod carries about 1.0 N along its length, 34 under its buckling load, and the deck moves 0.03 mm. To put a set in a rod you would have to push about 3.2 N sideways on its middle alone, or 1.5 N on the free tip of a glazing bar; nothing on a desk does the first, and the second is a fingernail hooked under the roof. The coaster is the fuse: the lighthouse tips, lands on brass, and stands up again.

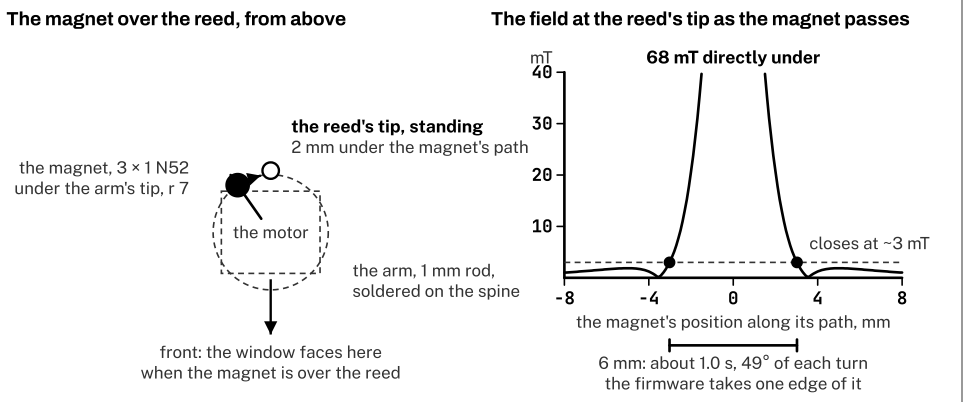


The lighthouse to scale with its centre of mass, with and without the washers, the line to the pivot at the coaster's edge, and the push at the ball that tips it. The rows are the fuse's numbers.

A magnet, a reed and one edge a turn

A reed switch is two ferromagnetic blades sealed in glass with a gap between them. A field along the glass magnetises the blades, their tips attract, and they touch: no supply, no current while open, two wires. A Hall sensor would want 3.3 V, three wires and a few milliamps all day, for a position the firmware needs once a revolution.

The magnet is a 3 × 1 mm N52 disc under the collar, 7 mm from the axis, and the reed stands on end under its path with 2 mm between. Straight under the magnet the field at the reed's tip is 68 mT, pointing down the glass, the direction a reed likes. A small magnet's field falls as the cube of the distance: 16 mT at 4 mm, 6 at 6, 1.4 at 10. So as the magnet comes round, the field at the tip rises from nothing to 68 mT and back inside a few millimetres of travel.



Left: the collar from above with the magnet coming round to the reed's tip; the window faces front when the magnet is over the reed. Right: the vertical field at the tip against the magnet's position (a dipole, so the middle is off the chart), the level a reed closes at, and the 6 mm of travel it stays closed for. The field reverses 3.5 mm out, which a reed does not mind.

A reed of this sensitivity closes at a few millitesla along it, 3 in this model, which the magnet's field reaches about 3.0 mm either side of centre: 6 mm of travel at 5.9 mm/s, a pulse of about 1.0 s, 49° of each turn. The pulse's length depends on which of the twenty reeds is picked and how it sits, so the firmware never uses it. It uses one edge, the same edge every turn, which lands at the same angle to a degree or two: the time between edges feeds the speed loop, and the park is that edge plus one offset written into flash on the first bench run, when the window is turned to face front by eye. The brake's 1.4° is the only scatter left.

The pull-up on GPIO6 draws 10 µA averaged over a turn. The glass is the one fragile part at the foot: the leads bend clear of the seal, and the tube is glued to the coaster on Kapton where the arm can never touch it.

The spine as a shaft, and a wheel for a washer

The spine is a 4 mm brass tube over the motor's 3 mm D shaft, held by a drop of gel glue, and it runs 144 mm from the motor's own bearing to the hub on the deck, a ring of the 5 mm tube from the same telescoping set, whose bore gives it about 0.1 mm of play all round. The shutter sits just above that bearing, so its top can wander at most 0.13 mm against 5.8 mm of clearance to the nearest rod. A bowed tube is the one thing that binds: roll it on the desk before cutting and swap a bowed one.

Rev C cut the shutter's floor, the magnet's collar and the deck's bushing from 0.4 mm sheet; Rev D makes them from 1 mm rod and tube rings because the builder does not fold sheet. The loads are tiny: the shutter is 5 g on two bars, and a 1 mm brass rod reaching from the spine to the drum's wall sags about 1 µm under its share, nothing a finger could feel. What matters is squareness, so the wheel is soldered up on the spine itself, hub ring first, spokes against its flanks, rim ring on their ends: the spine is the jig, and the bearing cannot be off centre. 63/37 solder melts at 183 °C, well under anything the brass minds.

Nothing in Rev D is painted or covered. The motor's steel can, the glass reed, the brass arm and the cell's black tray stand on the coaster in the open, each a colour the house rule allows, and there is no closed surface to lift. The lighthouse is brass, warm light and one dark cell.

The numbers on the sheet, and where they come from

NUMBER	WHAT IT IS
`Fl W 7.5s`	the light's character: 120° of 360° × 7.5 s = 2.5 s of light, 5 s of brass
`48°/s`	the shutter's speed: 360° / 7.5 s; the pool's edge crosses a wall 2 m away at 1.7 m/s
`19 lm`	everyday light: 250 mW × 110 lm/W × 0.7; 2.0 cd sideways (Φ / π²); 0.49 lux at 2 m, 2 moons
`84 mA`	the lamp at full: the filament's curve against (4.95 − V) / 23.5 Ω; the sheet's 83 with a flat 3.0 V
`134 / 46 mA`	the same lamp straight from the cell through 8 Ω, full and empty: why the rail is regulated
`52 %`	what the lamp gets of what the cell pays: 0.88 for the boost × 3.0 V of 5; a constant-current driver would save 37 mA
`43 %`	the duty for 8 rpm with a 0.8 V dead band (the sheet's 32 % has none); the reed loop finds it
`226 mA`	a stalled rotor and a cold lamp in one step: 5 V × 0.43 / 15 Ω + 83, over the boost's 200 mA start
`129 mA`	from the cell with the beacon on: 70 mA × 5 V / 0.88 / 3.7 V + 20 + 1
`3.55 V`	the cell above which the controller's rail is a true 3.3 V: 3.3 + 0.1 regulator + 0.15 diode
`650 mW`	heat in the charger: (5 − 3.7) V × 0.5 A; the coaster warms about 6 °C
`2.3 h`	the charge: 0.8 × 950 / 500 at constant current, then the taper; the 6 h timer is 2.6×
`35 N`	a rod's buckling load, fixed at both ends: π^2 E I / (0.5 L)^2; it carries 0.17
`47×`	the lean's gift to the sway stiffness: (2 E A $\sin^2\theta$ / L) over (48 E I / L^3), plus one
`0.3 N`	the push at the ball that tips it: 2.13 N × 37.5 / 243 mm; 0.5 with the washers
`68 mT`	the field at the reed's tip under the magnet: a 3 × 1 N52 disc at 2 mm, on its axis

`1.0 s`	the reed's pulse in the model: 6 mm of travel at 5.9 mm/s; the firmware uses one edge
`0.13 mm`	the shutter's wander: 0.1 mm of play × (1 + 37 / 144), against 5.8 to the nearest rod

Which part each idea lives in

- **a fixed lamp and a turning window, Fl W 7.5s, a third of the azimuth:** the 5/8 in brass tube, the shutter
- **speed from voltage, 1000:1, the dead band, the ramp into the boost:** the N20 gearmotor
- **a diode behind a resistor, the flat load line, 2700 K, 28 lumens:** the LED filament
- **a 5 V rail that does not sag, 88 %, the 200 mA start, power path, 650 mW:** the charger board
- **0.53 C, 90 % to the knee, calendar life at 4.2 V:** the Nitecore 16340 cell
- **the springs are the only resistance in the path:** the 16340 holder
- **the VBUS pin, the 3.3 V floor, 20 mA awake, a noisy ADC:** the ESP32-C3 SuperMini
- **two half-bridges, one of them a switch, flyback inside:** the MX1508 driver
- **0.15 V between the cell and a flashing cable:** the SS14 Schottky
- **the cube law, a one-second pulse, one edge:** the glass reed switch
- **1.45 T of remanence, 68 mT at 2 mm:** the N52 disc magnet
- **Euler at 190 to one, the tripod's 48×, 8 mΩ:** the 1.5 mm brass rod
- **a 3 mm D in a 3 mm bore, 0.2 mm in the bearing:** the 4 mm brass tube, the spine
- **80 g at the bottom, 25° and 31°, the fuse:** the brass coaster
- **two bars, an arm, a wheel: three sheet discs in rod and tube, 1 µm of sag:** the 1.0 mm brass rod

What comes next

Rev D, the open lantern with no house and its three small brass pieces made from rod and tube, awaits the builder's approval. Of the eight Amazon lines ordered on 2026-11-09 (five orders, $122.64; two boxes due 11-11, the filaments 11-27) the brass sheet is cancelled and the acrylic tube goes back; the five lamp parts from Grand Brass are ordered by the builder's word; the charger (1528-6106-ND, or Adafruit 6106) is still to buy, since that morning's DigiKey order was the rover's. When the parts come: measure the check ring's thickness, the vase cap's height and the coaster's hollow and rerun build_page.py; cut the jumper before the cell meets the charger; meter the tray's leads. The first bench run writes the park offset and the ADC's calibration point into flash. The first-power measurements the record wants replace this page's figures: the current from the cell in each mode, the lamp's current at full, the period from the reed, the charge current with the jumper cut.

Built by `docs/pages/gen/physics_page.py` from the dimensions, masses and budget in `build_page.py`; it also writes `docs/PHYSICS.md`. Assumptions, each a number at the top of the script: a 1000:1 gearbox, a 15 Ω winding with 1 mH, a 0.8 V dead band, a 30 ms brake; the filament at 3.0 V and 100 mA with 50 mV per e-fold and 1.5 Ω inside, 110 lm/W, 0.65 of its power as heat; 10 W/m²K off the lantern and the coaster; 0.1 V of regulator headroom; 0.8 of the charge at constant current; brass at 100 GPa and 200 MPa; an N52 disc at 1.45 T and a reed that closes at 3 mT. The charger's and boost's figures are TI's and Adafruit's; the motor's and the filament's are their listings. Nothing here has been measured on the lighthouse itself.